

Vinicius Santana

Software Engineer | Backend, Distributed Systems & Reliability

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SUMMARY

Software Engineer experienced in building Python/FastAPI services, maintaining Java/Spring systems, debugging production failures, measuring performance, and operating reliable PostgreSQL-backed platforms. Built validation, integration, and automation workflows for municipal healthcare, processing 21,000+ monthly records, reducing a 240+ person-hour reconciliation cycle to about 4 hours, and sustaining claim rejection rates below 1% for 8 consecutive cycles.

TECHNICAL SKILLS

Languages: Python, Java, SQL, TypeScript, Bash

Backend & APIs: FastAPI, Spring, Pydantic, SQLAlchemy, Alembic, REST APIs, OpenAPI, background workers, idempotency

Architecture & Security: Multi-tenant systems, authentication and authorization, JWT/OIDC, repository/adaptor architecture, event-driven workflows

Databases & Data: PostgreSQL, Firebird, DynamoDB, Redis, InfluxDB, Parquet, S3, AWS Glue, Athena

Testing & Quality: Pytest, pytest-asyncio, Hypothesis, Testcontainers, Ruff, Pyright, property-based and integration testing

DevOps & Observability: AWS (Lambda, SQS, SES, NAT Gateway, ELB), CDN, Cloudflare WAF, Docker, Linux, Git, GitHub, Jenkins, CI/CD, Ansible, Nginx, Prometheus, Grafana

PROFESSIONAL EXPERIENCE

Municipality of Presidente Epitácio - Presidente Epitácio, SP, Brazil | Nov 2021 - Present

Software Engineering & Health Data Systems (official title: Administrative Clerk) | Oct 2023 - Present

- Built Python/FastAPI validation services and maintained Java/Spring/PostgreSQL systems, modeling longitudinal patient identity across historical names, missing or changed CPF identifiers, and CNS changes for 21,000+ monthly records.
- Diagnosed Java/Spring performance degradation caused by an unset optional identifier that prolonged object lifetime and increased garbage-collection pressure; fixed identifier assignment and cleanup behavior to restore stable execution.
- Built performance instrumentation to detect N+1 patterns, repeated execution paths, and latency regressions, validating changes through baseline/diff comparisons, shadow runs, and night/chaos tests.
- Debugged an edge agent across installation, discovery, and execution using isolated VMs, MITM traffic inspection, and runtime diagnostics; traced repeat-install failures to stale Windows Registry and machine state, then updated uninstall cleanup and forced discovery during installation.
- Reduced healthcare claim rejection rates from 12%+ to below 1% across 8 consecutive billing cycles by engineering 11 auditable Python/SQL validation and reconciliation rules across legacy and government systems.
- Cut municipality-wide workforce reconciliation from 240+ person-hours to about 4 per cycle by building Python extraction, Parquet processing, provenance controls, and centralized APIs.
- Built an S3/Parquet analytical path cataloged by AWS Glue and queried through Athena, and worked on SQS, Lambda, and SES workflows for cross-application audit events and transactional email.
- Authored installation, operations, troubleshooting, and handoff guides, and trained 5 staff members plus 3+ physicians and nurses on production workflows, auditing, and issue resolution.

IT Infrastructure & Systems Support | Internship; promoted to role above | Nov 2021 - Oct 2023

- Designed and shipped a domain-specific plugin that reduced a mandatory dengue-reporting workflow from 7 manual steps to one checkbox.
- Operated health-check-driven failover for Nginx and application replicas across distributed municipal nodes, shifting traffic from unhealthy nodes within approximately 1–3 minutes with no reported user-visible outages.
- Improved release and recovery reliability with Ansible/Jenkins automation, health checks, backup checkpoints, certificate and credential rotation, and rollback procedures; restored a 9 GB PostgreSQL-backed application within 1 hour after a failed update.

Irmãos Santana - Panorama, SP, Brazil | Feb 2021 - Jan 2026

Business Analyst & Operations Manager | Family Business, Part-Time

- Increased profitability by 26% and reduced feed waste by 11% for a 20-ton/month aquaculture operation by redesigning pricing, purchasing, inventory, cost-control, and production workflows.
- Built Python decision-support tools, spreadsheet and ERP automations, and MQTT telemetry prototypes to measure production KPIs, inventory, water quality, and operating costs.

SELECTED PROJECTS

CnesData | 2024 - Present | Python, FastAPI, PostgreSQL, SQLAlchemy, Alembic, Parquet, S3, TypeScript, React

github.com/VINICIUS/CnesData

Research and portfolio implementation under development for a future external pilot; separate private municipal validation and reconciliation workflows operate in production.

- Designed and implemented a multi-tenant FastAPI ingestion architecture with PostgreSQL RLS, typed contracts, migrations, and adapters separating legacy sources from Parquet and S3 storage.
- Designed an AWS-native control plane using DynamoDB for tenant and job state, S3/Parquet with Glue and Athena for analytics, and SQS for decoupled background processing.
- Established CI quality and performance gates with unit, integration, property-based, negative, and containerized tests plus static analysis, microbenchmarks, and memory checks.

Limnopulse | 2026 - Present | Python, FastAPI, DynamoDB, InfluxDB, Redis, MQTT, AWS

github.com/VINICIUS/limnopulse

Portfolio platform under active development and testing; not yet in production.

- Designed and implemented tenant-authorized APIs for telemetry, metrics, alerting, and incident workflows using FastAPI, DynamoDB, InfluxDB, Redis, and MQTT.
- Kept incident state consistent across retries with optimistic concurrency, idempotent writes, and a durable outbox; validated concurrency, recovery, and duplicate-delivery behavior with automated tests.

e-SUS PEC Bootstrap | Jun 2026 - Present | Python, Docker, MinIO/S3, WAL-G, Prometheus, Grafana

github.com/VINICIUS/esus-pec-bootstrap

Used to provision and validate the municipal production e-SUS PEC environment described above.

- Built reproducible Python/Docker provisioning for e-SUS PEC across LXC/VM deployments, covering first-run configuration, TLS, Gov.br OAuth, data imports, monitoring, and health checks.
- Engineered and validated production backup, WAL archiving, restore, secret-handling, and rollback workflows, and documented recovery procedures for operators.

EDUCATION

B.Sc. in Computer Science - Universidade Paulista (UNIP) | Graduated Jul 2026 | GPA: 8.85/10.0

Information Technology - Systems Development Track - IFSP | Completed Dec 2021 | GPA: 8.87/10.0

LANGUAGES

Portuguese: Native | English: B2